

inverter for a pooled system would not cost 10X but closer to 3X. Also, 1kVA inverters would have power conversion efficiency of about 80 per cent, whereas a 10kVA inverter would have an efficiency of 95-98 per cent," explains Shankar. Thus there are cost and quality advantages to having a microgrid solution over independent systems (refer Fig. 2).

How it works

Ganesh and Hari explain the working by referring to Fig. 1. Basically, the entire system or solution consists of energy consumers (individual houses) and an energy generation setup. Individual houses are connected to the power network via a smart meter, which can communicate with the generation end. One of the main components used in this solution is the solar PV panel. It is placed on the rooftop of every house. Another vital component is the solar converter, which performs three tasks. It charges the battery from solar power, converts the DC power from solar or battery to AC, and if the microgrid is connected to electricity board (may be in the near future), then feeds the excess power to the main grid.

The battery incorporated in this solution stores the solar energy to power the house when there is no sunlight.



Fig. 4: Digital meter used for energy monitoring and communication at individual houses

The feature of Energy and Health Monitoring system monitors the plant operation and ensures the plant is operating in the limit of safety and gives an alarm if it is not in a safe health condition. This feature enables preventive maintenance of the plant. The most important aspect of this solution would be to tabulate the energy consumed by each individual house and also communicate the energy and health data to the remote location via Internet. For this there is a communication module that communicates with the individual meters. This helps the plant operator to virtually monitor the performance and health and also control the plant with an internet connection, while being located anywhere in the globe.

The smart meter can measure the electrical parameters and can also con-

nect or disconnect power according to the operator's instructions, and it can communicate with the communication module continuously. So the power and meter communication network forms the channel for power transfer from the

generating point to individual houses. The communication network could be wired or wireless based on the geographic spread.

In the future

Depending upon the geographical location of various other renewable energy sources such as wind, micro-hydel or biogas, this system can be integrated with solar as the primary source.

"If in case the electricity board extends the grid, then it can be fitted to the microgrid seamlessly. Diesel generator can also be integrated as a contingency plan (for power when there is emergency or during maintenance)," says Hari.

Ganesh informs, "The first version is ready. Based on the feedback we get from the initial takers, we will implement more features. A fully functional working prototype is ready in our lab and we are looking forward to work with government bodies, NGOs, international bodies, private setups and social enterprises in building as many smart renewable energy microgrids as possible."

Adding to that, Hari says, "We wish to sell the smart renewable energy microgrid tools to government, NGO or private setups who will operate the system using the tools we have developed for the operation of a microgrid. We will also sell them billing software, which will help them to collect appropriate charges from the individual power consumers of the microgrid." ●

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Why DC microgrid is less effective than AC microgrid

1. Household appliances like bulbs, fans and television sets are commonly available with AC power as the input, whereas getting the same in DC format is difficult. If one is still able to get a DC system, it will cost more than the AC system.
2. Also, AC products are more evolved, as they are manufactured by leading electronics companies. Though AC power is converted to DC in most systems, opening up a system just to trace the DC point is not feasible. DC utilities are not scalable in the Indian market.
3. Power dissipation (resistive loss) in AC wiring is much lesser than in DC wiring. For houses in the vicinity (in a DC microgrid), this could lead to reduced overall system efficiency. In our case study, we could do a calculation for both the systems and get the exact figures.
4. The efficiency of inverters increases with power capacity, while the cost reduces. For instance, one can get an inverter with 6kW power capacity and an efficiency of more than 90-95 per cent at three times the price of a 1kW inverter, which will have an efficiency of about 85 per cent. For a microgrid with more than ten houses, the peak power required would be about 6 kW. If those ten houses are powered with DC power, the cost and efficiency would perhaps converge with AC power.
5. Also, during some emergency or maintenance, a diesel generator set (an AC power source) can be connected to the system in order to ensure that the consumer is not deprived of power.

—Contributed by: Ganesh Shankar & Hari D.K.